5 Related Literature Review

The literature on pumped-storage hydropower addresses several related questions: whose value is assessed, which uncertainties affect operation, which markets provide revenue, and how the plant's capacities affect its value. This chapter reviews these questions in separate thematic subsections, in which the studies are compared, contrasted and critiqued, and connected to the present study. Section 5.1 defines the scope, Section 5.2 examines valuation perspectives, Section 5.3 discusses uncertainty and price behaviour, Section 5.4 examines market participation, and Section 5.5 considers capacity configurations. Section 5.6 then compares the solution approaches used in the literature and the physical detail they retain. Section 5.7 draws the findings together and identifies the research question motivating the present study.

5.1 Scope and Selection of Literature

The review focuses on operational scheduling and economic valuation of pumped-storage hydropower. Particular attention is given to studies that connect short-term operating opportunities with longer-term reservoir management. Studies of conventional reservoir hydropower are included where they offer relevant insights into water values, uncertainty or reserve markets. System-planning and hybrid-energy studies provide context for evaluating storage under different objectives, and studies of electricity-price behaviour are included because the research question concerns price volatility and persistence.

TODO (authors): Describe the search process you actually used: databases (e.g. Google Scholar, Scopus), search terms, inclusion criteria (peer-reviewed, relevance of title and abstract), and whether further studies were found by following reference lists (backward snowballing). Bore and Brensholm describe how they extended the review of their project report by backward snowballing. Three to five sentences are enough. The search log (`literature/search-log.md`) is still empty.

The chapter provides a focused narrative synthesis of selected studies. Their relevance is assessed through the economic question addressed, the representation of operation, the market opportunities considered and the treatment of plant capacities. Individual studies may appear in several subsections, but each discussion concerns the dimension relevant to that subsection. This allows the literature to be compared thematically while distinguishing findings that arise under different assumptions.

5.2 Valuation Perspective

The perspective adopted in a study determines what is meant by the value of pumped storage. System-oriented studies assess its contribution to the wider electricity system, whereas

owner-oriented studies assess the economic benefit available to an individual producer. The two answer different questions, and conclusions cannot be transferred directly between them.

Helseth et al. (2013) examine pumped storage within a hydrothermal system that also includes wind generation and transmission constraints, and Helseth et al. (2018) assess a pumped-storage expansion through changes in expected socio-economic surplus. Zhang et al. (2026) compare seasonal and conventional pumped storage in a Chinese system with much wind and solar power. The seasonal plant reduces curtailment, emissions and operating cost, but at higher capital cost and a higher levelised cost of electricity. Such results show what storage contributes to the system, but a favourable system-level assessment does not establish the corresponding private return, since benefits can also accrue to consumers and other producers. Likewise, additional owner revenue does not measure all the system benefits.

Owner-oriented studies instead evaluate the income associated with operating a particular asset. Ak et al. (2019) report revenue gains from operating an existing hydropower cascade in pumped-storage mode, and Alic et al. (2023) assess the economic benefit of adding a pump to a Norwegian cascade. Within an owner perspective, the treatment of market power introduces a further distinction. Abgottspon and Andersson (2013) consider a price-making producer whose trading decisions influence prices, whereas Helseth et al. (2016) consider a price-taking producer facing given prices. A price taker responds to available price differences, while a price maker also accounts for how its own actions reduce them.

Increased operating revenue alone does not establish investment profitability. Bore and Brensholm (2026) take this further step and use additional revenues to estimate the maximum investment cost that a capacity upgrade can support, which requires an economic lifetime, a discount rate and cost assumptions. Ak et al. (2019) make a simpler check by comparing the annual revenue gain with the annuity of the extra investment in reversible turbines. Conclusions about the economic value of storage must therefore be read against the objective of the study from which they are drawn.

5.3 Uncertainty in Pumped-Storage Scheduling

Uncertainty affects the trade-off between using stored water now and retaining it for future operation. For pumped storage, future prices influence both the opportunity to sell electricity and the cost of replenishing storage through pumping. Natural inflows provide an additional uncertain source of water in inflow-dependent plants. The literature differs in which of these uncertainties is represented, how they are described over time, and what information is available when operating decisions are made.

5.3.1 Sources of Uncertainty and Their Dependence

Electricity prices are central to the owner-oriented studies reviewed here. Alic et al. (2023) account for uncertainty in both prices and inflows in a pumped-storage cascade, and Bore and Brensholm (2026) likewise consider uncertain inflows and prices for conventional reservoir hydropower. Helseth et al. (2016) additionally treat reserve-capacity prices as uncertain. Other uncertain inputs become relevant when pumped storage is studied as part of a wider

energy system. Naval and Yusta (2025) consider wind uncertainty in the operation of a hybrid system, whereas Zisos et al. (2023) examine uncertainty in wind, demand and energy conversion in an island-energy design problem. In both, prices play a subordinate role: the first treats a full year of day-ahead prices as known, and the second contains no prices at all. Inflow assumptions appropriate to a reservoir-fed cascade also need to be distinguished from those of a closed-loop plant without inflow, such as the one studied by Chazarra et al. (2017).

When several inputs are uncertain, their dependence becomes a modelling choice. Kleiven et al. (2023) find that ignoring the negative correlation between Norwegian prices and inflow lowers water values and leads to a higher reservoir in autumn, but changes expected revenue very little. Helseth et al. (2016) treat prices as independent of inflow and justify this only qualitatively, and Pérez-Díaz et al. (2020) find the correlation between energy and reserve prices too weak to affect water values for a Spanish plant. These findings suggest that dependence matters more for how the reservoir is operated than for the revenue it earns. They concern conventional plants, however, and do not show whether the same holds when prices also drive pumping.

5.3.2 Timing and Resolution of Information

Long-term storage decisions depend on the short-term operating opportunities available when the water is eventually used. Seasonal inflows and prices govern how the reservoir is managed over the year, whereas price variations within the day and week create the opportunities for pumping and generation. Ak et al. (2019) illustrate the division: when their cascade is operated as pumped storage, the monthly reservoir decisions barely change, while the allocation of pumping and generation over the hours of the day changes considerably. A long evaluation horizon therefore does not by itself establish that a study captures storage flexibility.

The timing of information distinguishes studies even when they include the same uncertain variables. Bore and Brensholm (2026) represent uncertain conditions between weeks but assume that the current week's inputs are known when its decisions are made. Their model therefore combines future uncertainty with perfect information within each week. An hourly operating schedule does not, by itself, imply that uncertainty is resolved hour by hour. Helseth et al. (2013) make the same weekly assumption and note themselves that it is too coarse to capture the balancing needs created by wind power.

Abgottspon and Andersson (2014) examine when this simplification matters. For a Swiss pumped-storage plant, they compare four ways of representing operation within the week, from an aggregated model to one in which information arrives within the week. Treating the week as known overvalues water. A stochastic representation of the week gives about 10% higher expected profit, and a similar improvement in the poor outcomes, but only when weekly secondary reserves are included; without reserves, the model with a known week performs better. The authors conclude that stochastic within-week detail is worthwhile when the market structure is sufficiently complex, and suggest that complex plant structures may be another reason, but leave this untested. The result rests on a single plant whose capacities are

not reported, and profits are evaluated with an operator's dispatch heuristic, so differences between the models mix policy quality with the heuristic's behaviour.

Time resolution and information are related but separate choices. Abgottspon and Andersson (2016) show that, for a Swiss system with seasonal storage reservoirs and small balancing reservoirs, a purely daily model undervalues and a purely hourly model overvalues the plant by more than 10%, whereas linking daily and hourly decisions comes within a few per cent. Their reported case is deterministic, however. Alic et al. (2023) find that simulating operation day by day rather than week by week lowers revenue considerably in weeks when a state-dependent environmental restriction becomes active, because the weekly model knows the reservoir state too coarsely. Testing several price scenarios can establish sensitivity to different conditions, but it does not by itself establish how decisions adapt as information arrives.

5.3.3 Representing Uncertainty over Time

Uncertainty is typically represented by a small number of weekly states derived from data, and the reviewed studies share several steps. Historical or simulated series are collected, grouped into a few typical states per week, and transition probabilities between states are estimated. Alic et al. (2023) cluster weekly prices and inflows into five joint states, and Bore and Brensholm (2026) into three. Helseth et al. (2016) build nine states from three energy-price and three reserve-price clusters of only 80 simulated scenarios, and have to adjust the transition probabilities slightly because so few scenarios are available for nine states. Mo et al. (2001) describe how such probabilities can be corrected so that the conditional expected prices are preserved. The number of states is thus constrained by the data: more states describe the variation better but rest on fewer observations each.

Aggregation within the state affects what the pump can earn. Alic et al. (2023) represent each state by weekly average prices, spread over the week with a fixed daily profile, so the price variation within the week is the same in every state. Bore and Brensholm (2026) use the median series of each cluster, and Ak et al. (2019) represent hourly prices through representative daily price curves for each month, which removes the chronology of pumping and generation. For pumped storage, whose income comes largely from short price differences, such averaging may understate the value of pumping. This may help explain why Alic et al. (2023) find a small gain from their pump, although the authors do not make this argument. An alternative is to use historical years directly. Helseth et al. (2018) let each historical weather and price year occur with equal probability and correct it to the current state with a simple autoregressive model, and Helseth et al. (2013) simulate operation on historical records to retain correlations that their stochastic models omit. Data granularity should therefore be chosen with the source of the plant's income in mind.

5.3.4 Price Volatility and Persistence

A statement that a study “includes uncertainty” is not sufficient to characterise it. For a storage owner trading on price differences, the representation of price behaviour over time is likely to be at least as important as the number of uncertain inputs. Two properties of this

behaviour are central: volatility, how much prices vary around their expected pattern, and persistence, how long high- or low-price conditions last (Section ??).

Haldrup and Nielsen (2006) find that Nordic day-ahead prices are highly persistent, most of all in the Norwegian price areas, which they link to the slow variation of reservoirs and inflow. Their data are from 2000–2003, however, before market coupling and the growth of wind and solar power, and their regimes last hours rather than weeks. A weekly state description captures persistence only through its transition probabilities, and Mo et al. (2001) note that such a description forgets its initial state at an exponential rate, so uncertainty about prices far ahead needs an additional model. Future price behaviour may also change: Henden et al. (2016) find that more transmission to the continent increases the variability of southern Norwegian prices, whereas more pumped storage reduces it.

Studies that relate price behaviour to storage value do so mostly with known prices. Ak et al. (2019) find the largest gain from pumping in the price year with the largest variation within the day, although the relation is not one-to-one. Bradbury et al. (2014) find that higher volatility raises the return on storage in seven US markets. Sioshansi et al. (2009) show that a simple rule based on the prices of the previous two weeks captures most of the perfect-foresight arbitrage value, because daily and weekly price patterns repeat. Short-term variation also matters more for pumped storage than for conventional hydropower: Löhndorf and Wozabal (2023) find that coordinating day-ahead and intraday trading is worth about an order of magnitude more for a pumped-storage plant than for a reservoir plant without a pump, consistent with the small value of intraday trading that Faria and Fleten (2009) find for a conventional cascade. None of these studies varies volatility and persistence separately or relates them to the value of pump, turbine and reservoir capacity under uncertainty. Since pumping and generating power earn from short price differences, whereas reservoir capacity earns from differences that last over weeks or seasons, the two properties may favour different capacities.

5.4 Electricity Market Participation

Market participation determines the revenue opportunities available to a storage owner. Energy-market participation rewards electricity sales relative to the cost of purchases for pumping. Other products can remunerate the availability of operating flexibility or its actual use. Studies must therefore be compared by the products and settlements included, since these define what their reported revenues cover. The reported gains are not directly comparable either, as market design, price levels, plants and comparison baselines differ between the studies.

Energy and forward trading

Energy-centred studies provide a baseline for assessing the value of pumping. Ak et al. (2019) and Alic et al. (2023) value pumping against day-ahead prices only, and Alic et al. (2023) add environmental restrictions that limit how fast the reservoir level may change. Abgottspon and Andersson (2013) add forward contracts for a price-making Swiss producer. Physical production and trading positions are then separated: in one simulated week, a forward

obligation is met by buying cheaper energy back in the day-ahead market rather than by generating. Forward trading raises simulated annual profit by about 20%. The result depends on the price-making assumption, with a price response estimated from experience, and although the study is described as stochastic in prices, only inflow is represented as random. Forward positions extend the producer's commercial opportunities, but they serve a different purpose from payments for maintaining reserve availability.

Reserve capacity

Reserve participation changes scheduling and water values as well as revenue. Helseth et al. (2016) compare energy-only scheduling with joint sales of energy and symmetric reserve capacity, traded the week before energy, for a Norwegian watercourse of seven plants. A large part of the reserve income is earned without changing the generation schedule, because the plants are most efficient below full output and therefore have spare capacity in both directions. Where the schedule does change, the large reservoirs are kept higher until the spring flood and then used more to keep generators running in the low-load season, some of them at minimum output. In their case, water values are lower with reserves, since capacity is withheld from generation in periods of high output. Helseth et al. (2017) find a similar pattern for a single reservoir in southern Norway: selling primary reserves raises expected operating profit by 7 to 11%, shifts less energy between seasons, and makes water values fall faster as the reservoir fills. Both studies evaluate their policies on scenarios drawn from the same model used to compute them, and the second fits its price model to only three years of data.

Reserve commitments compete with other uses of the same flexibility. In Helseth et al. (2017), energy revenue falls when reserves are sold, so the net gain is smaller than the reserve income. Chazarra et al. (2017) find that a closed-loop plant earns most of its income from secondary reserve capacity, and that the most flexible configuration accepts a loss in the day-ahead market in order to offer reserves in every hour. Pérez-Díaz et al. (2020) report a gain of about 4.5% from including the automatic frequency restoration reserve market when computing water values for a Spanish plant, and find that accounting for its own effect on reserve prices reduces spillage considerably while barely changing profit. Bore and Brensholm (2026) find that capacity markets make up a larger share of revenue in periods of high inflow and low energy prices, and that attractive capacity prices can make the plant spill water and restrain production to keep capacity available for upward reserves. Reserve income must therefore be evaluated together with the operating opportunities given up to provide it.

A pump widens what a plant can offer. A conventional plant cannot provide reserves while pumping, which is why the configurations of Chazarra et al. (2017) that can pump and generate at the same time earn more. Dallinger et al. (2019) show that adding pump-turbines to an Austrian reservoir plant extends its operating range to between about −70% and 100% of turbine capacity, so it can offer downward reserve without generating and no longer depends on natural inflow to deliver reserves. Abgottspon and Andersson (2014) include weekly secondary reserves for a Swiss pumped-storage plant and find that they raise expected profit in all four of their models.

Reserve activation

The treatment of activation further distinguishes market representations. Chazarra et al. (2017) include regulation energy associated with upward and downward reserve use, based on historical activation, and show that storage limits restrict reserve offers: the conventional plant does not offer reserves in an hour in which the expected activation could empty its upper reservoir. Their results rest on two representative weeks of Spanish data with prices and activation known in advance. Dallinger et al. (2019) simulate activations every quarter hour but offer capacity at a price of zero, so their reserve revenue comes from activated energy only. Bore and Brensholm (2026) include day-ahead and intraday energy alongside FCR-N, FCR-D and upward and downward aFRR capacity, but exclude activation. Compared with Chazarra et al. (2017), this shows that the number of included products and the completeness of their delivery representation are separate dimensions of market coverage.

Consequently, studies of market participation need to distinguish energy trading, reserve-capacity commitments and reserve activation. Revenues from these activities interact because they depend on shared operating capability. Findings from different market combinations cannot be compared solely through their total income, particularly when their treatment of activation and settlement differs. Most of the multi-market evidence also concerns conventional reservoir hydropower. For pumped storage, only Abgottspon and Andersson (2014) combine reserves with the long-term value of water, for a plant whose capacities are not reported.

5.5 Pumping, Generating and Reservoir Capacities

Capacity studies differ in which plant characteristics are fixed and which are allowed to change. Optimising the operation of an existing plant, comparing specified upgrades and optimising the plant's dimensions address different questions. An effective operating policy for one configuration does not establish whether that configuration is well sized.

Adding a pump

The most concrete configuration comparison in the literature is the addition of a pump to an existing cascade, and the reported gains are modest. Ak et al. (2019) find that operating a Turkish cascade as pumped storage raises turbine revenue by a quarter to a third, but net revenue by only about 6% on average after the cost of pumping. Total net energy supplied falls, most of the gain comes from the reservoir with the largest storage, and the most downstream plant loses revenue because no water is pumped into it. Alic et al. (2023) add a 38 MW pump to a Norwegian cascade and find that median energy production rises by about 9% but median revenue by only about 2%. The production figure counts gross turbine output, so recirculated water is counted twice, and only the revenue reflects the cost of pumping. Their pump also has the same power as the existing turbine, so the effect of pump size cannot be separated from that of adding a pump at all. Schüller et al. (2024) value a new plant with pumping in the Sira-Kvina system and find that the gain is smaller when operation is simulated with hourly detail than in the medium-term model, because short-term restrictions such as a minimum head for pumping limit the use of the new plant. All three studies use the energy market only.

Upgrading generating units

Helseth et al. (2017) compare two reinvestment alternatives of the same total generating power. The configuration with two Pelton units gains more from reserve sales than the one with a large Francis unit, because Pelton units can run at low output and leave more room for reserves, although the Francis unit has the higher peak efficiency. The value of a configuration thus depends on the markets included. Bore and Brensholm (2026) examine generating-capacity upgrades of 30, 60 and 90 MW for a conventional plant. Energy revenue hardly changes with the upgrade, because the plant has no more water to sell, while capacity-market revenue grows substantially, and the marginal benefit of each step falls. Additional generating power is thus limited by the water available, and its value depends on markets that pay for capacity rather than energy. Their alternatives concern generating power only, in a plant without a pump.

Large expansions and market conditions

Large capacity additions can change the prices they rely on. Helseth et al. (2018) expand the 200 MW Duge pumped-storage plant by up to 2000 MW in a model of the Nordic power system, in which prices follow from the system operation. A detailed representation of the watercourse gives a lower marginal value of the additional capacity than an aggregated model, because the detailed constraints of the surrounding reservoirs limit its use. The paper does not state whether pump and turbine are expanded separately, and the pumped-storage case relies on an assumed exchange capacity to Germany chosen to create price differences that trigger pumping. Henden et al. (2016) find that a large expansion of pumped storage in Norway removes much of the price differences it trades on and leaves its owners with little income, and Dallinger et al. (2019) find that the price spread captured by Austrian pumped storage falls as more storage is built. An expansion of this scale is therefore valued differently from an upgrade evaluated under given prices.

Sizing the reservoir

Design optimisation brings selected dimensions into the decision problem itself. Zisos et al. (2023) optimise the upper reservoir and the photovoltaic capacity of an island system while holding pumping and generating power fixed at 6 MW each. Their benchmark optimum is a reservoir of about 315 000 m^3, less than a third of an earlier proposal of 1.1 Mm^3, although the authors caution that the two rest on different assumptions. Across weather and demand scenarios, the scenarios that call for a deeper reservoir give lower profit, and the authors use this relation to show the range of profit a designer accepts by choosing a larger, more conservative reservoir. The plant follows a fixed rule that pumps all surplus and covers all deficits, without prices, and each scenario has its own optimal design, so the study does not identify one design that performs well across scenarios. Evidence from electricity markets points to diminishing returns from additional storage duration. Sioshansi et al. (2009) find that most of the arbitrage value of a storage plant is earned by its first hours of storage, and Bradbury et al. (2014) find that the most profitable ratio of storage to power is set mainly by efficiency and by how costs divide between power and storage. Both studies use known prices and only energy arbitrage. Chazarra et al. (2017) further observe that their most flexible

configuration uses only a small part of its reservoir, and suggest, without testing it, that this could affect reservoir sizing.

The reviewed studies therefore provide evidence on operating given assets, introducing pumping, comparing expansions and selecting reservoir dimensions. Together they indicate what limits the value of a capacity: the water available, the cost of pumping, short-term operating restrictions, the markets that pay for flexibility, and the effect of the plant itself on prices. They differ in which capacities are varied and which remain fixed. This matters because an additional unit of one capacity may have limited value when another capacity is binding; a larger pump, for example, offers little additional storage opportunity when the reservoir is already full. None of the reviewed studies varies pumping power, generating power and reservoir capacity jointly for a plant that trades in electricity markets.

5.6 Solution Approaches for Stochastic Hydropower Scheduling

The reviewed studies use different approaches to connect current operation with uncertain future opportunities. Comparing them requires distinguishing the formulation of the decision problem, the algorithm used to solve it, and the procedure for updating decisions over time. The principles are introduced in Sections ??, ?? and ??; this section compares how they have been applied and what each application had to give up.

Physical representation

Physical operating detail affects the mathematical problem to be solved, and the studies differ widely in how they represent pumping, generation, head and efficiency. Chazarra et al. (2017) use a mixed-integer formulation to represent the operating modes of each unit, including hydraulic short-circuit operation, in which water is pumped and generated at the same time. With ternary units this mode more than doubles the income of their closed-loop plant, whereas with reversible units the gain is small. Ak et al. (2019) use a nonlinear model with head-dependent generation but the same efficiency curve for pumping and generation. Helseth et al. (2017) represent head-dependent generation curves and unit commitment with integer variables. Several studies instead assume constant head and efficiency (Alic et al., 2023; Naval and Yusta, 2025; Zisos et al., 2023), and Helseth et al. (2018) treat head as a parameter updated each week. These choices determine which operating behaviour a model permits: a continuous model allows low output and small reserve adjustments that real units may not deliver, and a constant efficiency makes every pumping cycle equally profitable. Their effect on conclusions has been measured for profit, but not for the ranking of plant configurations.

5.6.1 Two-Stage Stochastic Programming

Two-stage formulations distinguish decisions taken before uncertainty is resolved from subsequent decisions that depend on its realisation. A second-stage scenario can cover many operating periods, even though it represents only one further information stage. Faria and Fleten (2009) use a two-stage stochastic mixed-integer programme for the day-ahead bidding of a Norwegian cascade, with intraday trading as the second stage, and find that anticipating

intraday trading changes profit very little. Helseth et al. (2018) use two-stage stochastic linear problems for detailed hydrothermal scheduling of the Nordic system, with scenarios for weather and exogenous prices resolved in the second stage and solved by Benders decomposition. This retains the full hydropower system without aggregation, but simplifies the future information structure: each scenario is known once the first week has passed, so the plan for later weeks is made with foresight. Head is represented as a parameter with precomputed sensitivities rather than as a variable, so the linear formulation does not capture non-convex operating relationships.

5.6.2 Rolling-Horizon Planning

Rolling-horizon planning updates an operating plan repeatedly as new information becomes available. The initial decisions are implemented, the system state is updated, and the problem is solved again. This procedure can be applied to deterministic or stochastic formulations. Helseth et al. (2018) embed their two-stage problems in a rolling-horizon simulator that moves one week at a time. To keep this tractable, they reduce the number of historical scenarios, use a coarser time resolution further into the future, and take the value of water at the end of the scenarios from another model; even so, a simulation takes 50 to 156 hours. Deterministic rolling schemes are cheaper. Sioshansi et al. (2009) optimise two weeks at a time with a 15-day horizon so that stored energy keeps a value at the end of each window, and Dallinger et al. (2019) optimise each week with a two-week look-ahead. Re-optimisation introduces adaptation between successive solves, while each individual solve retains its assumed information structure. Such approaches should therefore be compared with a multistage formulation on the basis of how decisions can adapt, rather than on the total simulation horizon alone.

5.6.3 Stochastic Dynamic Programming

Stochastic dynamic programming (SDP) connects current storage with future operating opportunities through a value function computed on a grid of reservoir levels, and places no convexity requirement on the stage problems. Alic et al. (2023) choose SDP because their plant is subject to a state-dependent water-abstraction restriction, which is non-convex, alongside minimum environmental flows and limits on how fast the reservoir level may change. The restrictions matter: the ramping limits move production from high-price to lower-price periods and reduce revenue, and a finer grid improves accuracy at a computational cost of several hours for two reservoirs. Helseth et al. (2017) combine SDP with weekly mixed-integer problems. A linear version of their model overestimates profit with reserve sales by 2 to 5 percentage points, mainly because it lets units sell reserves below their minimum output. Linear water values used in a mixed-integer simulation lose only about one percentage point, which suggests that much of the linearisation error arises in the operation within the week rather than in the water values. Abgottspon and Andersson (2014) use SDP between weeks together with hourly subproblems within each week, and keep the small daily reservoir out of the state by resetting it every week. Pérez-Díaz et al. (2020) propose SDP as a benchmark for quantifying the approximation errors of linear policies.

The flexibility of SDP comes at a computational cost. The number of storage-state combinations grows rapidly as reservoirs are added, and the reviewed applications accordingly concern one plant or a small cascade. Its suitability depends on the size of the state space as well as on the benefit of retaining non-convex operating detail.

5.6.4 Stochastic Dual Dynamic Programming

Stochastic dual dynamic programming (SDDP), introduced by Pereira and Pinto (1991), approximates future value functions using cuts, avoiding a full grid over reservoir volumes, and is regarded as the state-of-the-art method for medium- and long-term scheduling in hydropower-dominated systems (Helseth et al., 2016). The convergence analysis of Shapiro (2011) concerns multistage linear stochastic problems, and standard cut arguments do not extend directly to non-convex formulations. Applications therefore approximate the physical detail that SDP can retain. Helseth et al. (2013) include pump start-up costs through a continuous operating state instead of integer variables. This reduces switching from generation to pumping sharply, but the pump still runs below its intended minimum output, so the approximation captures the effect of start-up costs without providing an exact commitment schedule. In the same study, transmission constraints and a finer, chronological time resolution change cable use and the assessed profitability of the cable, and raise the run time several times over. Helseth et al. (2016) combine SDP for the price process with SDDP for the reservoirs, which allows inflows, energy prices and reserve prices to be uncertain, at a run time of more than two days, and warn that the linear plant description may cause significant errors when reserves are sold.

Extending the method

Several studies trade computation time for physical accuracy. Hjelmeland et al. (2019) include head-dependent generation and use strengthened cuts, which give a better operating policy at about five times the run time, although the gap between the bounds remains large when prices are uncertain. Helseth (2024) connect an aggregated representation of the Norwegian hydropower system with feasibility cuts computed beforehand from the detailed system. This lowers total cost compared with plain aggregation at almost three times the run time, and is considerably faster than the higher-dimensional cuts of the earlier method of Helseth and Mo (2023). Such extensions narrow the gap between the two dynamic-programming approaches without removing the underlying trade-off.

Applications to individual owners

Applications of SDDP to an individual owner's valuation remain concentrated on conventional reservoir hydropower. Bore and Brensholm (2026) use SDDP with weekly decision stages and a linear generation–discharge relationship. Each iteration samples one path through the weeks, the algorithm stops after a fixed number of iterations, and no bounds are reported; they attribute a water value that does not fall as the reservoir fills to the limited range of reservoir levels visited. For pumped storage, Abgottspon and Andersson (2016) use SDDP in a multi-horizon model but report a deterministic case, and Schüller et al. (2024) use water values from an SDDP-based model to value a pumping investment in the energy market.

Owner-oriented pumped-storage studies that include reserve markets have relied on SDP or deterministic optimisation.

The reviewed approaches consequently offer different combinations of computational tractability, operational detail and adaptation to new information. Two-stage formulations simplify the future information structure; rolling-horizon planning repeatedly updates decisions; discretised SDP can retain non-convex stage problems for small systems; and SDDP exploits a convex structure to approximate future value for larger ones. Their relative suitability follows from the decision problem and the assumptions retained, rather than from a universal ranking of the methods.

5.7 Synthesis and Research Gap

Table 3 summarises selected studies along the dimensions discussed above. The literature establishes several relevant strands: economic valuation from different perspectives, representation of uncertainty, participation in multiple markets, comparison of plant configurations, and approaches for solving the resulting scheduling problems. These strands address complementary aspects of the value of pumped-storage flexibility.

The closest studies cover different parts of the capacity-valuation question, and each provides a different piece of evidence. A system study shows that a large pumped-storage expansion has a lower marginal value once the watercourse is represented in detail (Helseth et al., 2018). Owner studies show that adding a pump yields modest gains in the energy market and that its value is limited by pumping costs and short-term restrictions (Ak et al., 2019; Alic et al., 2023; Schüller et al., 2024). Studies of reserve markets show that the most valuable configuration is the one with the widest operating range (Helseth et al., 2017; Chazarra et al., 2017), and an island study shows that reservoir size involves a trade-off between cost and reliability (Zisos et al., 2023). These studies differ in objectives, uncertainty, horizons and markets, as they were designed to answer different questions. Their findings therefore cannot be combined into a common assessment of pump, turbine and reservoir capacities for a pumped-storage owner.

Bore and Brensholm (2026) already connect stochastic multi-market operation to the value of generating-capacity upgrades. The remaining research need concerns the joint contribution of pumping, generating and storage capacity in a pumped-storage setting. It also concerns the influence of price dynamics: the reviewed studies relate price behaviour to storage value only with known prices, and none separates volatility from persistence, although the two may favour different capacities.

Against this background, the present study asks how pumping power, generating power and reservoir capacity, individually and in combination, affect the owner's long-term operating value across electricity markets. A related question is how their relative value changes with price volatility and persistence. The intended contribution is a comparative understanding of these capacity interactions under uncertainty, building on the scheduling and valuation approaches established in the reviewed literature. The last row of Table 3 places the study alongside the reviewed literature.

Table 3: Selected studies compared by valuation perspective, modelling approach, market participation and capacity question, and the positioning of this study.

Study	Perspective and asset	Time and uncertainty	Model representation	Market focus	Capacity question
Abgottspon and Andersson (2013)	Price-making owner; Swiss pumped storage	One year; inflow	SDP with hourly recourse	DA and forward contracts	Operation of the studied plant
Abgottspon and Andersson (2014)	Owner; Swiss pumped storage	One year; inflow and prices; known or uncertain week compared	SDP with stochastic or simplified within-week models	DA and weekly secondary reserve capacity	Operation of the studied plant
Helseth et al. (2016)	Price-taking owner; reservoir hydro	Medium term; inflow, energy and reserve prices	Combined SDP/SDDP; linear	Sequential energy and reserve-capacity markets	Water values and schedules with reserve access
Helseth et al. (2017)	Price-taking owner; reservoir hydro	One year; weekly states of inflow and prices	SDP with weekly MILP; head-dependent generation	DA energy and primary reserve capacity	Two reinvestment alternatives
Chazarra et al. (2017)	Price-taking owner; closed-loop pumped storage	Daily scheduling; known prices and activation	MILP; hydraulic short-circuit operation	DA, secondary reserve and regulation energy	Three unit configurations
Ak et al. (2019)	Owner; cascade with pumping	Annual evaluation; historical inflows and price years	NLP with monthly decisions and operating rules	DA energy	Existing cascade versus pumped-storage operation
Alic et al. (2023)	Owner; pumped storage in a cascade	One year; weekly states of inflow and energy prices	SDP; state-dependent environmental constraints	DA energy	Adding a 38 MW pump; effect of constraints
Zisos et al. (2023)	Hybrid island-energy design	Long-run simulation; wind, demand and conversion	Rule-based simulation with heuristic search	Supply reliability and economic performance	Reservoir and PV sizing; fixed pump and turbine power
Schüller et al. (2024)	Price-taking owner; Norwegian watercourse	Inflow and prices	SDDP-based water values with hourly simulation	Energy	A new plant with pumping
Naval and Yusta (2025)	Hybrid-system operator; pumped storage	Hourly operation over a year; wind; prices known	MILP with chance constraints	Grid exchange and renewable demand coverage	Operation of a specified hybrid configuration
Bore and Brensholm (2026)	Price-taking owner; reservoir hydro	One year; weekly stages; inflow and prices; known within-week inputs	SDDP; linear generation model	DA, ID, FCR-N, FCR-D and aFRR capacity; activation excluded	Generating-power upgrades; maximum feasible investment cost
This study	**Price-taking owner; pumped storage with natural inflow**	**Weekly stages; inflow and DA prices; known within-week inputs; varied price volatility and persistence**	**SDDP; linear plant model**	**DA energy and reserve capacity; activation excluded**	**Combinations of pumping power, generating power and reservoir capacity**

SDP: stochastic dynamic programming; SDDP: stochastic dual dynamic programming; NLP: nonlinear programming; MILP: mixed-integer linear programming; PV: photovoltaic; DA/ID: day-ahead/intraday; FCR: frequency containment reserve (N: normal operation, D: disturbances); aFRR: automatic frequency restoration reserve. System-perspective studies and general methodological references are discussed in the text but are not included in this focused comparison.

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